

DAY-5

Tasks

1.

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

```
C:\Users\SIDDHI>ping 192.168.31.83

Pinging 192.168.31.83 with 32 bytes of data:
Reply from 192.168.31.83: bytes=32 time=456ms TTL=64
Reply from 192.168.31.83: bytes=32 time=2ms TTL=64
Reply from 192.168.31.83: bytes=32 time=17ms TTL=64
Reply from 192.168.31.83: bytes=32 time=14ms TTL=64

Ping statistics for 192.168.31.83:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 456ms, Average = 122ms
```

2.

Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
C:\Users\SIDDHI>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4826:400::]
over a maximum of 30 hops:

  1     2 ms     1 ms     1 ms    jiofiber.local.html [2409:4090:1047:a819:caa3:e8ff:fe6a:e10f]
  2     *         *         *      Request timed out.
  3    11 ms    10 ms    10 ms    2405:203:400:100:172:31:3:218
  4    17 ms   108 ms    72 ms    2405:200:801:b00::f3c
  5    10 ms    11 ms     8 ms    2405:200:809:1510:61::6
  6     *         *         *      Request timed out.
  7    24 ms    21 ms    11 ms    2405:200:801:b00::dfe
  8    63 ms    73 ms   240 ms    2405:200:801:b00::dfd
  9    23 ms    28 ms    20 ms    2001:4860:1:1::1b8e
 10   181 ms   18 ms    20 ms    2001:4860:1:1::1b8e
 11    17 ms   14 ms    32 ms    2001:4860:4826:400::

Trace complete.
```

3.

Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```
C:\Users\SIDDHI>ping 192.168.31.83

Pinging 192.168.31.83 with 32 bytes of data:
Request timed out.
```

4.

Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Interface: 192.168.52.1 --- 0xb

Internet Address	Physical Address	Type
192.168.52.254	00-50-56-f0-29-37	dynamic
192.168.52.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.56.1 --- 0xd

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.11.1 --- 0xf

Internet Address	Physical Address	Type
192.168.11.254	00-50-56-f7-74-9d	dynamic
192.168.11.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static

224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.31.53 --- 0x12

Internet Address	Physical Address	Type
192.168.31.1	c8-a3-e8-6a-e1-0f	dynamic
192.168.31.254	f8-94-c2-9e-8f-a4	dynamic
192.168.31.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.102.18	01-00-5e-7f-66-12	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

5.

Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

Network Connectivity and Traceroute Report

I connected my laptop and my friend's phone to the same home Wi-Fi network. I first checked the IP address of the phone from its Wi-Fi network settings. The phone's IP address was 192.168.31.83.

I then opened Command Prompt on my Windows laptop and used the ping command to test connectivity between the two devices. I entered:

```
ping 192.168.31.83
```

The result was Ping statistics for 192.168.31.83:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 456ms, Average = 122ms.

This showed that the laptop and phone were able to communicate using ICMP ping.

Next, I used the `tracert` command to check the path from my laptop to YouTube:

```
tracert www.youtube.com
```

The traceroute completed after 11 hops. The first hop was my home router, whose IP address was 192.168.31.1. The remaining hops represented devices/routers between my home network and the destination.

I then disconnected my friend's phone from the Wi-Fi and ran the `ping` command again using its previous IP address. The result was Request timed out. This occurred because the phone was no longer connected to the network.

Finally, I used:

```
arp -a
```

to view IP addresses and MAC addresses known to my laptop. I identified the router, laptop and other available devices on the network and recorded them in a table.

Overall, the practical demonstrated how `ping` can be used to test network connectivity, `tracert` can show the path taken by network traffic, and `arp -a` can display IP-to-MAC address mappings on the local network.